

Checkpoint 5: Full Results

Objective

Evaluate the three main models on the grouped 384-image test split across three seeds.

Key Message

Every final quantitative claim uses the same grouped test list and reports mean +/- sample SD.

Completed Work

- Evaluated 128x128 50-epoch baseline and residual models.
- Evaluated default, parameter-matched, and residual models across three seeds.
- Generated grouped aggregate metric summary.

Verification and Controls

- Ran all three architectures with seeds 42, 123, and 2026 on the identical grouped test list.
- Aggregated seed-level scores as mean +/- sample standard deviation rather than pooling test images across runs.
- Cross-checked tables, plots, and qualitative examples against the machine-readable grouped summary.

Evidence

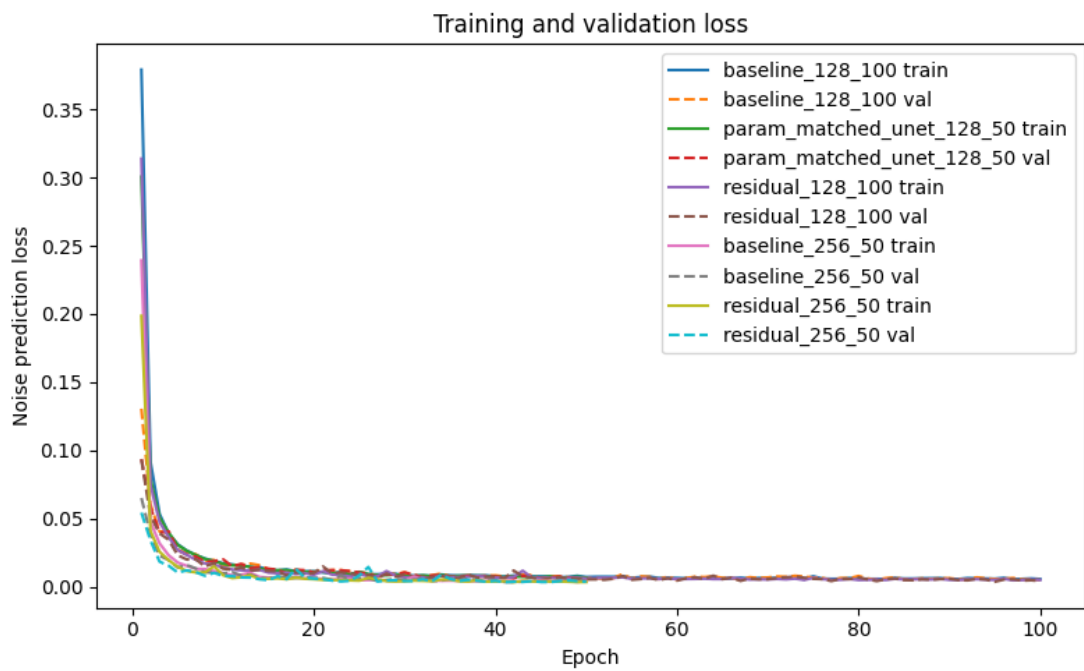
Experiment	PSNR mean +/- SD	SSIM mean +/- SD	Delta E mean +/- SD
U-Net	15.780514 +/- 0.187075	0.657180 +/- 0.015721	28.380208 +/- 1.320879
Matched U-Net	16.017719 +/- 0.277610	0.722221 +/- 0.022096	27.590478 +/- 2.280969
Residual	16.006089 +/- 0.267086	0.779948 +/- 0.006607	27.053617 +/- 1.081249

Decision / Next Gate

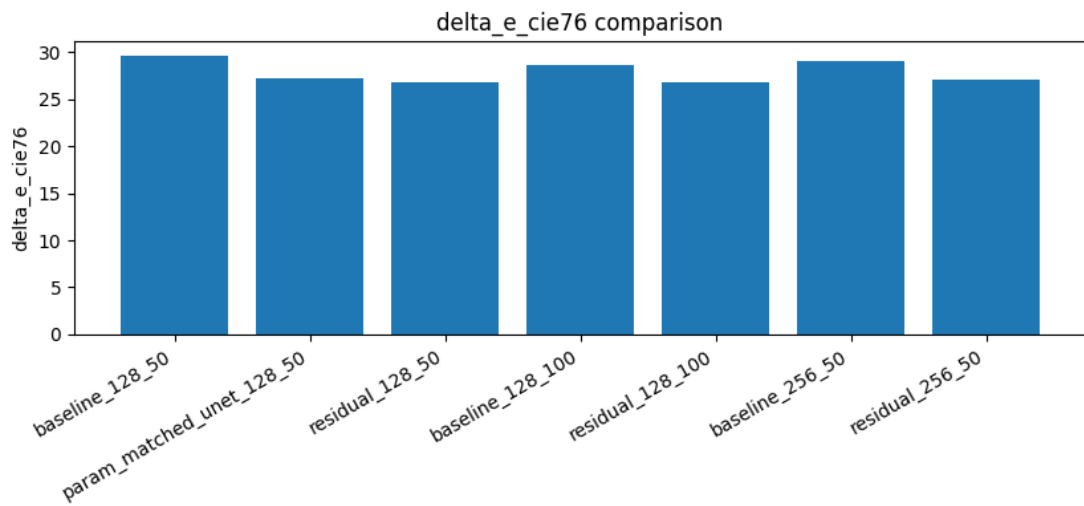
Use only the grouped three-model by three-seed matrix for final quantitative claims and keep entropy descriptive.

Notes

- Earlier duration and resolution ablations used the superseded split and are excluded from final claims.



Training and validation loss curves.



Delta E comparison across experiments.